NVIDIA DSX POD Reference Design

November 2026

NVIDIA is an AI Infrastructure Company

AI as multi-layer infrastructure driving global job creation and economic growth

APPLICATIONS

MODELS

INFRASTRUCTURE

CHIPS

ENERGY

CHATBOTS

DIGITAL BIOLOGY

ROBOTAXI

ENTERPRISE AI AGENTS

SCIENCE

ROBOTICS

MANUFACTURING

AI CODER

LLM	VLM	VLA	MMLLM	GPT
DM	GNN	MOE	SSM	LBM



NVIDIA

NVIDIA Vera Rubin Platform

Five rack-scale systems work as one AI supercomputer

NVIDIA Vera Rubin NVL72
NVLink spine



NVIDIA Groq 3 LPX
Direct Chip-to-Chip spine



NVIDIA Vera CPU
Spectrum-X Ethernet spine



NVIDIA BlueField-4 STX Storage
Spectrum-X Ethernet spine



NVIDIA Spectrum-6 SPX



NVIDIA

AI Factories Build the Intelligence Economy

Turn power and data into intelligence at industrial scale



AI factories turn power and data into intelligence

Standardization makes deployment faster and more repeatable

Faster deployment brings intelligence online sooner

Traditional Data Center Construction Delays AI Capacity

Complex on-site integration and skilled-labor shortages extend schedules and delay time to first token

288,000

Projected construction-worker shortfall by 2027

30-40%

Of project labor hours are electrical

600,000

On-site labor hours for one conventional facility

~300

Skilled workers on-site at peak build out phase

Labor constraints extend schedules while increasing cost and risk, constraining how quickly new compute facilities can come online

NVIDIA

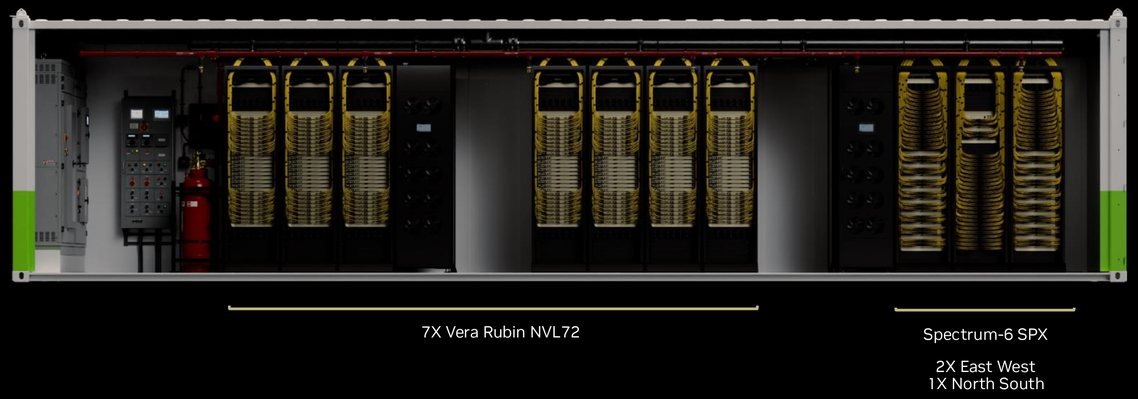
NVIDIA DSX POD

NVIDIA reference design for containerized AI factories

Factory integration accelerates time to first token
Vera Rubin infrastructure is installed, cabled, piped, and tested in the factory, reducing on-site work

On-site deployment can shrink from months to days
Connect containers to prepared site power, cooling, and networking, then complete bring-up and acceptance testing

Designed by NVIDIA, delivered by partners
NVIDIA engineers and validates the design, partners customize, manufacture, and support solutions

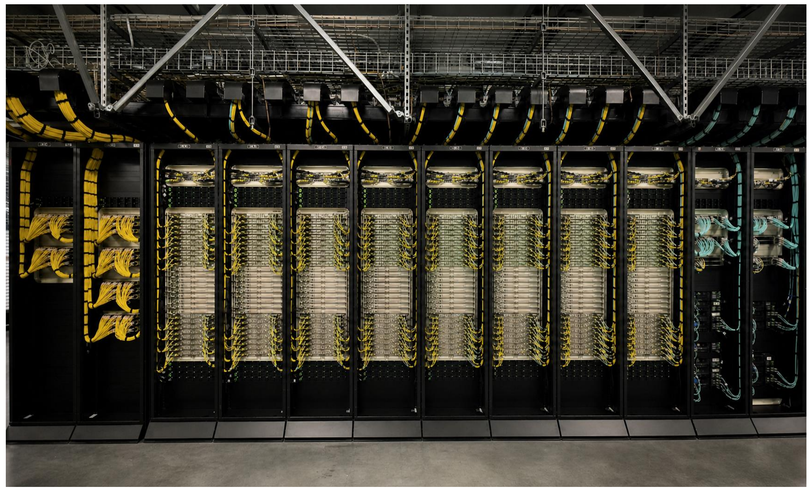


NVIDIA DSX POD Compute Container

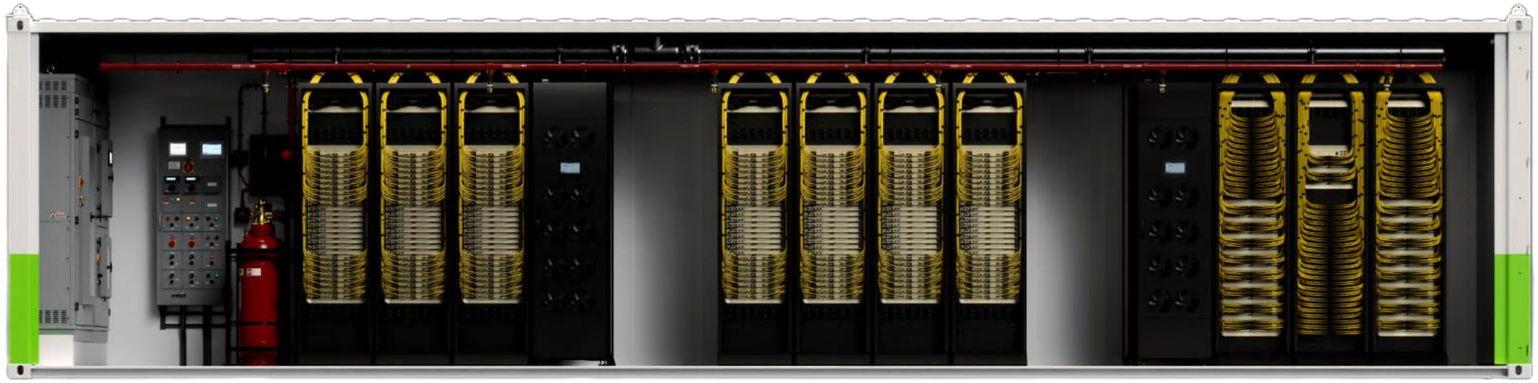
THE FASTEST PATH TO PRODUCTIVE INTELLIGENCE AT SCALE

Factory Integration Shifts Deployment from Months to Days

Factory-installed and tested infrastructure is ready to connect when the customer site is ready



Traditional Data Center Build



NVIDIA DSX POD Containers

6 months

- Install racks, systems, cabling, wiring, and piping on site
- Test after installation, fix issues, retest, repeat
- Skilled labor availability and site conditions can delay schedule

14 Days

- Containers arrive pre-integrated, cabled, piped, and factory-tested
- Connect to prepared site power, cooling, and networking
- Complete bring-up and final acceptance testing

Working estimates for on-site deployment at a fully prepared site. Timing varies by configuration and scope; excludes factory production, transport, permitting, and site construction.

NVIDIA

From Factory Delivery to Productive AI Capacity

Factory-integrated containers arrive ready for connection at prepared customer sites

01. Position

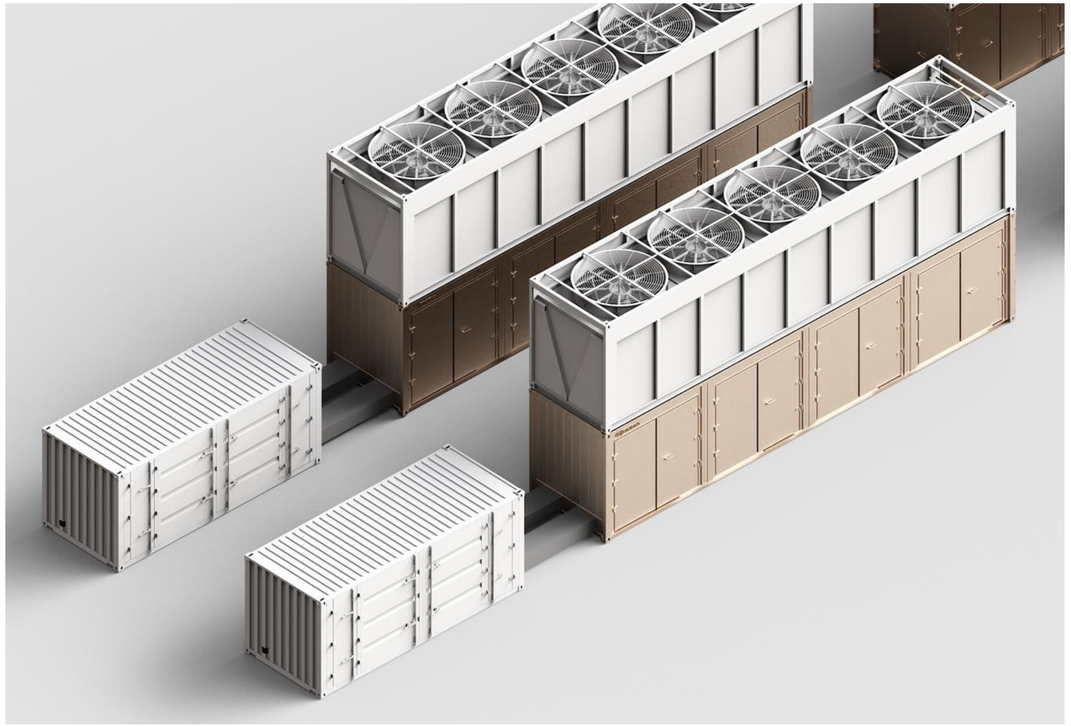
Deliver and position containers with systems installed and factory-tested

02. Connect

Connect to prepared site power, cooling, and networking

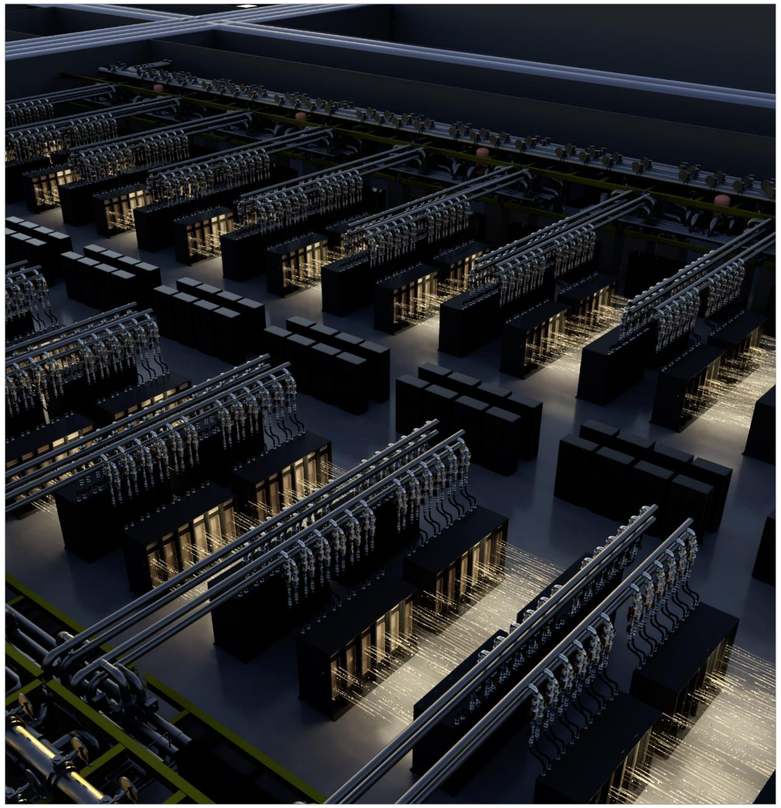
03. Bring Online

Complete system bring-up and final acceptance testing



NVIDIA DSX POD Designs Deliver Efficiency with NVIDIA DSX MaxLPS

Dynamic power management inside the AI factory to maximize throughput



Key Benefits

Policy-Driven Power Control

Dynamically enforces power events and policies to keep the data center's power usage within defined limits.

Dynamic Workload Orchestration

Ensures that compute jobs are scheduled and executed within the available power budget.

Compliance and Validation

Provides real-time, high-fidelity telemetry to validate that power targets are being met and to enable fast response.

Designed by NVIDIA, Delivered by Partners

Partners build and deliver solutions using the NVIDIA DSX POD reference design

Broad ecosystem of partners who:



Engineer and manufacture customer solutions



Reduce deployment risk with a validated design and modular expertise



Provide services and support

DSX POD Partners

ANDURIL
ARMADA
Bull
Crusoe
DAYONE
DELL Technologies
DELTA
EATON
GIGABYTE
HPE
ingrasys
Lenovo
PEGATRON
QCT
Schneider Electric
VERTIV
wistron

Preliminary Technical Specifications

Per DSX POD compute container, subject to change

Vera Rubin NVL72 Compute Container

Form Factor	ISO 40' HC container
Container Power	1.2 MW (MaxLPS); 1.4 MW (MaxQ); 1.7 MW (MaxP), 800V DC hard wired, 2.7 MW capacity
Container Cooling Interface	NPS10 (DN250) pipe
Thermal Solution	Chiller/dry cooler
Networking Interface	Trunk fiber@288F or 980F
Total components	>10M
Total weight	>28,500 KG
In Container	
NVL72 racks	7x Vera Rubin NVL72
Networking Racks	Spectrum-6 SPX, 2x E-W racks + 1x N-S rack
NVFP4 Inference	25.2 EFLOPS

NVIDIA


NVIDIA